Routing Models for Local VLMs on the SciVer Task

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SciVer is a multimodal chart-claim verification task

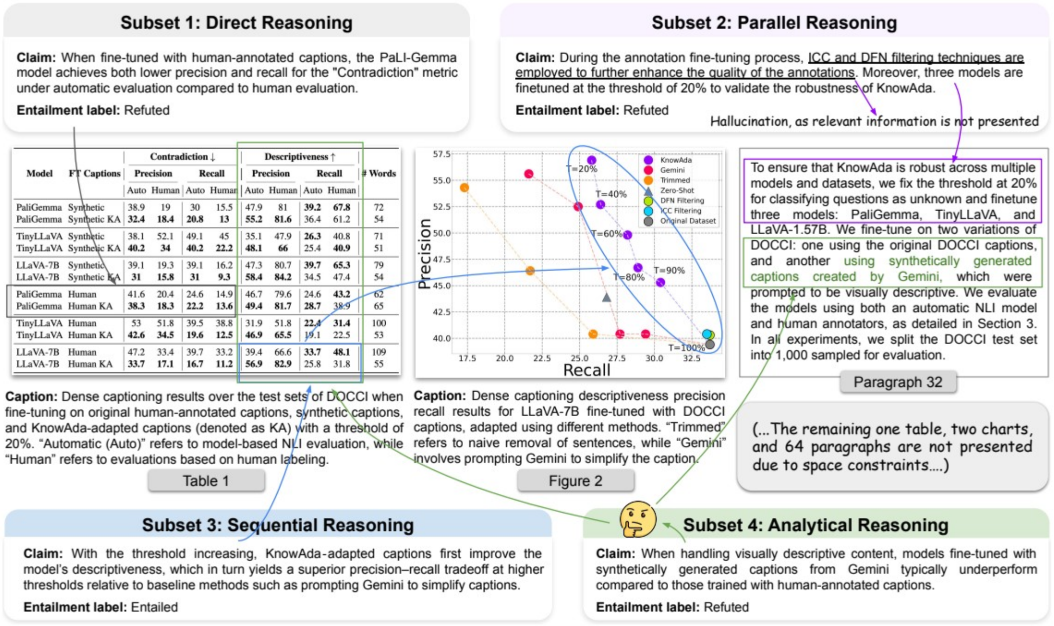


Figure 1: An illustration of the four subsets in the SciVer benchmark. Our benchmark is designed to evaluate document-grounded scientific claim verification in a multimodal setting. To effectively perform this task, models must go through the full context of a scientific paper—including text, charts, and tables—to locate the appropriate supporting evidence before verifying a claim. The complete data examples are provided in Appendix C.

Why this task motivates routing

- Each example pairs a scientific claim with multimodal paper evidence: text, charts, and tables.
- SciVer distinguishes direct, parallel, sequential, and analytical reasoning subsets.
- The original paper reports best average test accuracy of 77.7% for o4-mini versus 93.8% human expert accuracy.
- A router can preprocess each chart-claim pair and assign it to the VLM predicted to perform best.

Source: Wang et al., SciVer, arXiv:2506.15569, Figure 1 and Table 3. The same paper reports GPT-4.1 at 70.8% on analytical reasoning versus 90.0% for human experts.

The router workflow separates selection from evaluation

- 1. Build the matched, sanitized three-VLM router dataset.
- 2. Construct base, chart-TDA, claim-NLP, and combined feature sets.
- 3. Split by paper_id into grouped train/validation/test sets.
- 4. Select hyperparameters by grouped 10-fold CV using either mean-regret-first or top-1-first ranking.
- 5. Refit the selected hyperparameters on train+validation and evaluate once on grouped test.
- 6. Train a final all-data model for future use after evaluation artifacts are written.

router split	rows	paper groups
train	568	434
validation	123	95
test	120	90
grouped CV	10 folds inside train	paper_id kept intact
bootstrap	1,000 held-out resamples	resampled with replacement
seed	215	

Key guardrail: paper_id groups stay intact through train/validation/test and through the 10-fold CV inside the training split.

Key definitions for router evaluation

Observed five-run accuracy

fraction of five local runs that match the SciVer label for each item and VLM; target labels and regret are noisy when VLMs tie.

Router target

train one regression target per VLM accuracy, then select the VLM with the highest predicted accuracy.

Top-1 hit rate

fraction of held-out examples where the routed VLM is one of the observed best VLMs, counting ties as hits.

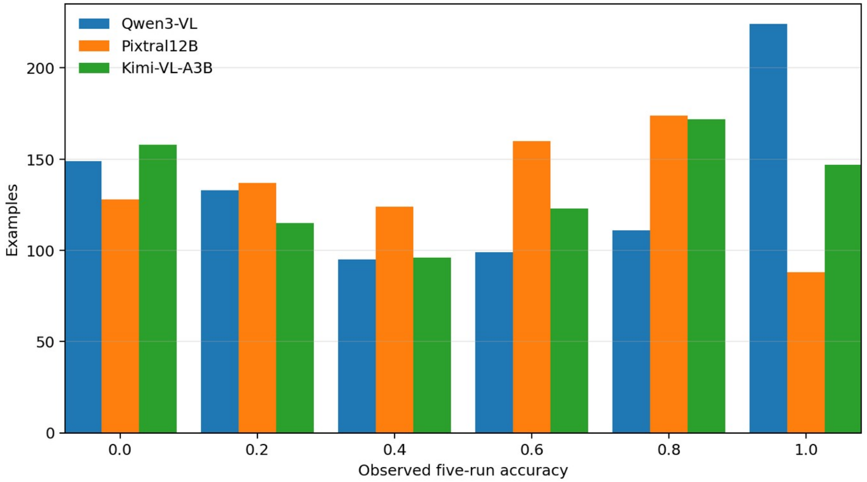
Regret

observed best accuracy minus the selected VLM's observed accuracy. Lower is better; zero means the router picked a best VLM.

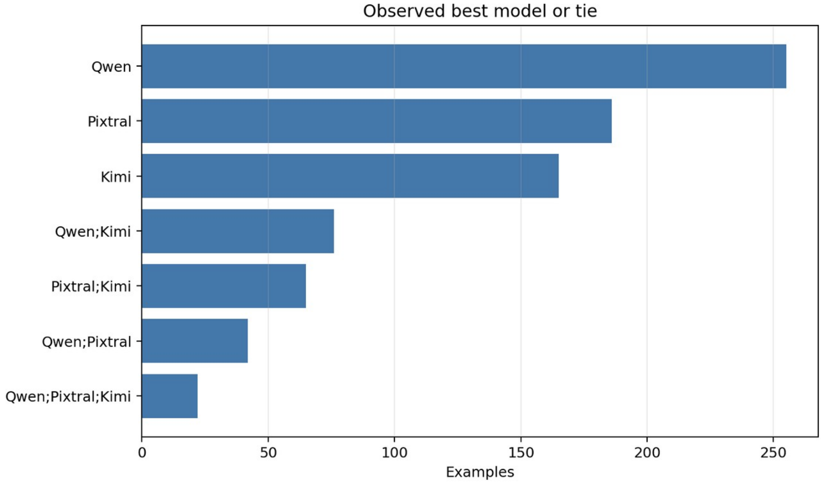
Grouped 10-fold CV

training paper groups are split into 10 folds; each candidate trains on 9 folds, predicts the held-out fold, and every paper_id stays in one fold.

Close VLM performance makes ties and switching matter



Because each item has five local repeats, observed accuracy can only take the six discrete values [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]. Each group of bars shows how many chart-claim items fall at that accuracy value for each VLM.



Counts show which VLM or tie-set has the highest observed five-run accuracy for an item. Best-model targets are recomputed only over Qwen3-VL, Pixtral12B, and Kimi-VL-A3B.

VLM	mean observed accuracy
Qwen3-VL	53.9%
Pixtral12B	49.3%
Kimi-VL-A3B	51.8%

Features combine difficulty, chart structure, and claim semantics

feature family	public columns	meaning
12 cognitive dimensions	12	Human-coded chart difficulty annotations included as precomputed public features; deployment would require these annotations or an automatic proxy.
Base scalar features	14	Simple size, token, numeric, and image-shape summaries.
Chart TDA	86	Topological summaries of grayscale and edge-map chart structure.
Claim NLP	81	TF-IDF/SVD semantics, cue counts, alignment, numeric overlap, and token-level TDA.

Interpretation

The model does not see raw claims, captions, prompts, responses, labels, or images. It sees sanitized derived columns that are available to reproduce router training.

Three router families test different assumptions about the feature signal

Ridge/ElasticNet	linear shrinkage	stable on small tabular data
XGBoost	boosted decision trees	captures nonlinear feature interactions
PyTorch MLP	one/two hidden layers	tests a small neural tabular router

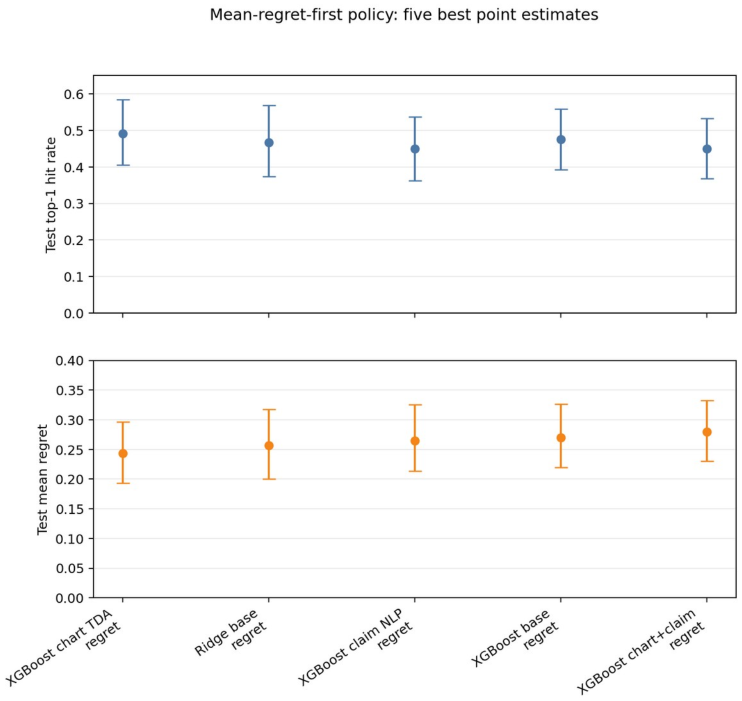
All three predict one observed five-run accuracy per VLM and route to the highest predicted value.

Evaluation: XGBoost + chart TDA leads the point estimates

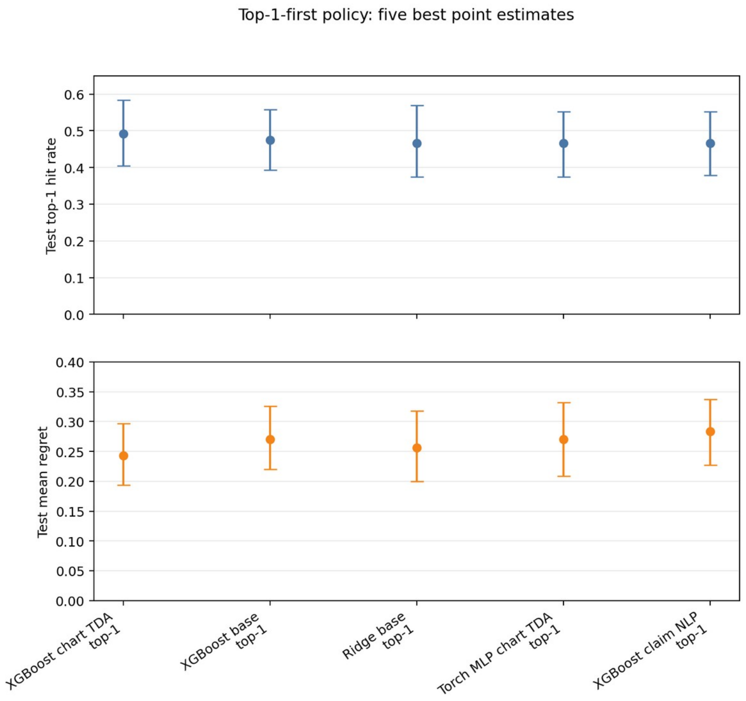
section	model / baseline	router	features	top-1	mean regret	median	selected candidate / models
Regret first	XGBoost chart TDA regret first	XGBoost	base + TDA	49.2%	0.243	0.200	XGB: 200 trees, depth 3, lr 0.05
Regret first	Ridge/ElasticNet base regret first	Ridge/ElasticNet	base	46.7%	0.257	0.200	Ridge alpha 0.1
Top-1 first	XGBoost chart TDA top-1 first	XGBoost	base + TDA	49.2%	0.243	0.200	XGB: 200 trees, depth 3, lr 0.05
Top-1 first	Ridge/ElasticNet base top-1 first	Ridge/ElasticNet	base	46.7%	0.257	0.200	Ridge alpha 0.1
Baseline	always pixtral12b	deterministic	-	41.7%	0.302	-	pixtral12b: 120
Baseline	always qwen3vl	deterministic	-	40.8%	0.305	-	qwen3vl: 120
Baseline	always kimi vl a3b	deterministic	-	40.8%	0.315	-	kimi_vl_a3b: 120
Baseline	random	stochastic	-	38.3%	0.295	-	kimi_vl_a3b: 38, pixtral12b: 38, qwen3vl: 44

The table combines the top two regret-first runs, top two top-1-first runs, and all baselines for the best-run split except train-global-best.

Bootstrap intervals show the top runs are promising but uncertain



Only the five best point-estimate runs under the mean-regret-first policy are shown, sorted by lower held-out mean regret. Top panel shows tie-aware top-1; bottom panel shows mean regret.

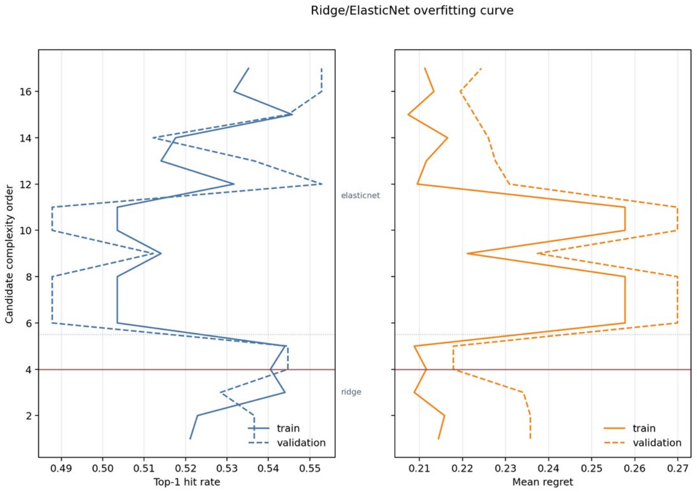


Only the five best point-estimate runs under the top-1-first policy are shown, sorted by higher held-out top-1 hit rate. Intervals are grouped-bootstrap intervals over held-out paper groups.

Intervals resample held-out test paper groups only, after hyperparameters are fixed.

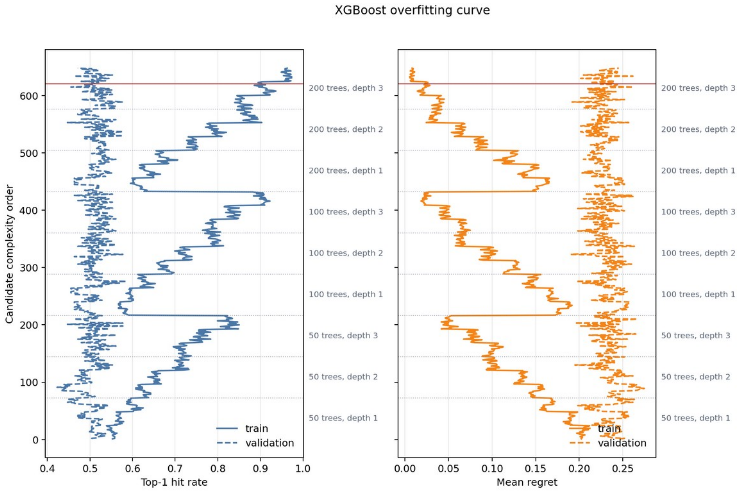
Overfitting diagnostics compare train and validation behavior

Ridge/ElasticNet



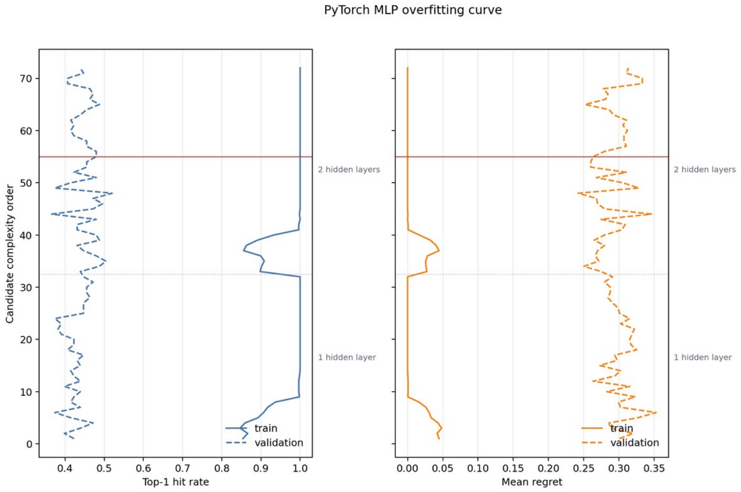
Curves show train versus grouped-validation top-1 and mean regret across regularization candidates for the best regret-first Ridge/ElasticNet run. The solid red horizontal line marks the selected candidate. Dotted horizontal separators mark Ridge versus ElasticNet candidate zones when both appear.

XGBoost



Curves show train versus grouped-validation metrics across the XGBoost grid. Dotted horizontal separators split tree-count/depth zones such as 50 trees depth 1, 50 trees depth 2, and so on; the right-side labels mark those zones. The solid red line marks the selected candidate.

PyTorch MLP



Curves show train versus grouped-validation metrics across the PyTorch MLP grid. Dotted horizontal separators split one-hidden-layer and two-hidden-layer candidates; the solid red line marks the selected candidate.

Reading: large train-validation gaps are warning signs. The selected XGBoost and PyTorch MLP candidates fit the training groups much better than validation groups.

Selected-candidate overfitting summary

router family	selected run	selected candidate	train top-1	validation top-1	top-1 gap	train regret	validation regret	regret gap	reading
Ridge/ElasticNet	Ridge/ElasticNet base	Ridge alpha 0.1	54.0%	54.5%	-0.4%	0.212	0.218	0.006	limited overfitting signal
XGBoost	XGBoost chart TDA	XGB: 200 trees, depth 3, lr 0.05	90.3%	51.2%	39.1%	0.023	0.231	0.208	strong overfitting signal
PyTorch MLP	PyTorch MLP chart TDA + claim NLP	MLP 32x16, tanh, alpha 0.01	100.0%	48.0%	52.0%	0.000	0.263	0.263	strong overfitting signal

Takeaway

Ridge/ElasticNet shows limited overfitting signal. XGBoost and the PyTorch MLP show strong training gains that do not fully carry to grouped validation, so bootstrap/test evidence is essential.

Acknowledgments and references

Acknowledgment

The 12 cognitive-difficulty dimensions used by this router study were engineered by the project AI Faithfulness in Scientific Reasoning.

Project repository: https://github.com/Erdos-Projects/summer26-ai-faithfulness-in-scientific-reasoning

SciVer paper and dataset

Wang et al., SciVer: Evaluating Foundation Models for Multimodal Scientific Claim Verification

Paper: https://arxiv.org/abs/2506.15569

Dataset: https://huggingface.co/datasets/chengyewang/SciVer

These sources provide the benchmark task, released dataset, and cognitive-difficulty annotation framework on which the routing experiments build.